Object Oriented Programming
Lab Assignment 3

22CS02009
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To Run:

1. Please place all the program files in a folder called “Lab_3”, this is because they were created under package name Lab_3, not doing so will give error.
2. To run program, go outside the Lab_3 directory and (# means number)
   a. `cd Lab_3`
   b. `javac *.java`
   c. `cd ..`
   d. `java Lab_#.Q#`

Question 1:

All Classes were created as required by the question.

```
Name: Snehal
Age: 20

Name: Atharva
Age: 21
Salary: 125000.0

Name: Aditya
Age: 21
Salary: 200000.0
Department: CP
```

Question 2:

All Classes were created as required by the question.

```
Descriptor: Company
Date: 1/1/1
Current Value: 10.0
Number of Shares: 100
Share Price: 6.7
Asset Type: stock
```

```
Descriptor: Government
Date: 9/7/1945
Current Value: 3.14
Interest Rate: 16.64
Asset Type: bond

Descriptor: Personal
Date: 1/1/2020
Current Value: 5000.0
Interest Rate: 5.5
Asset Type: saving
```

Question 3:

All Classes were created as required by the question, with mathematical formulas used accordingly.

```
Enter length, breadth, height of Box: 1 2 3
Surface Area of Box: 22.0
Volume of Box: 6.0

Enter length of Cube: 4
Surface Area of Cube: 96.0
Volume of Cube: 64.0

Enter radius and height of Cylinder: 1 2
Surface Area of Cylinder: 18.84955592153876
Volume of Cylinder: 2.0

Enter radius and height of Cone: 1 2
Surface Area of Cone: 10.166407384630519
Volume of Cone: 0.6666666666666666
```

Question 4:

All Classes were created as required by the question.

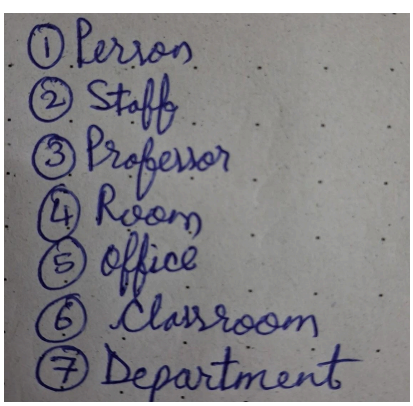
1. For Linked List based implementation,
    a. Class Node was created much like a C based struct based Linked List, the Java class based Linked List works the same, with no handling for freeing the Nodes, due to automatic garbage collection.
    b. It was ensured that enqueue and dequeue operations take O(1) time.
2. For Array Based Implementation,
    a. An array of maximum size 100 was defined, since an array cannot have its length changed dynamically at runtime unlike ArrayLists.
    b. It was ensured that enqueue and dequeue operations take O(1) time, by making the pointers for front and rear, and handling the shifting via cyclicity.

```
Linked List Queue:
Queue: 1 2
After enqueing 3, Queue: 1 2 3
After dequeing, Front: 2
Queue: 2 3

Array Queue:
Queue: 4 5
After enqueing 6, Queue: 4 5 6
After dequeing, Front: 5
Queue: 5 6
kumar-snehal@Kumar-Snehal-PC:~/OOP$
```

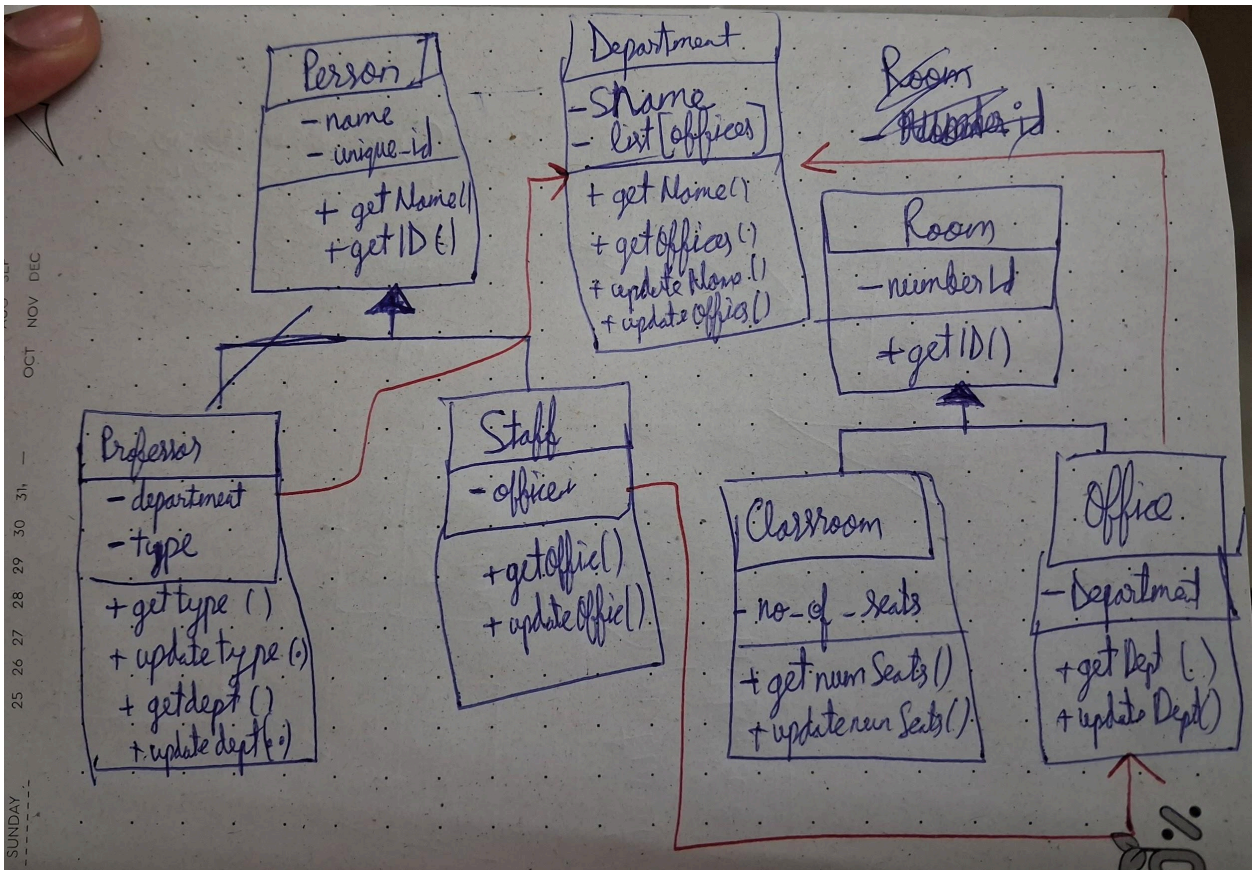
Question 5:

List of Classes (part a):



Relationship Diagram (parts b, c and d):

1. The data types are used according to the data as Integer, String or Enum for professor type.
2. Please note that the red lines refer to the referencing, here instead of using IDs, directly the object's reference is used by the classes in my code, this ensures that an imaginary classroom/ department/person won't be referred since it would be null.
3. The ids are given automatically via a constructor ensuring no repeatedness.




```
Departments:
Department Name: Computer Science and Engineering, Number of Offices: 3
Department Name: Electrical Engineering, Number of Offices: 2
Department Name: Electronics and Communication Engineering, Number of Offices: 1

Offices:
Number ID: 1, Department: Computer Science and Engineering
Number ID: 2, Department: Computer Science and Engineering
Number ID: 3, Department: Computer Science and Engineering
Number ID: 4, Department: Electrical Engineering
Number ID: 5, Department: Electrical Engineering
Number ID: 6, Department: Electronics and Communication Engineering
```

```
Classrooms:
Number ID: 7, Number of Seats: 60
Number ID: 8, Number of Seats: 120
Number ID: 9, Number of Seats: 240

Professors:
Unique ID: 1, Name: Prof. Alice, Type: full, Department: Computer Science and Engineering
Unique ID: 2, Name: Dr. Bob, Type: assistant, Department: Electrical Engineering
Unique ID: 3, Name: Dr. Charlie, Type: associate, Department: Electronics and Communication Engineering
Unique ID: 4, Name: Dr. Delta, Type: associate, Department: Computer Science and Engineering
Unique ID: 5, Name: Dr. Echo, Type: assistant, Department: Electrical Engineering

Staff:
Unique ID: 6, Name: Mr. Frank, Office ID: 1
Unique ID: 7, Name: Ms. Grace, Office ID: 4
Unique ID: 8, Name: Mr. Henry, Office ID: 6
Unique ID: 9, Name: Ms. Irene, Office ID: 2
kumar-snehal@Kumar-Snehal-PC:~/OOP$
```